

Lecture Notes: Signal & Transformations

Course: 2110203 Computer Engineering Mathematics II

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1 Introduction

1.1 Definition

A signal is a function that represents or conveys information about a physical phenomenon. It is typically expressed as a function of one or more independent variables. For example, a time-varying electrical current or voltage can be represented as a function of time. Other common examples of signals include speech, television images, temperature measurements, and light intensity.

1.2 Digital Signal Processing Applications

Digital Signal Processing refers to the use of digital algorithms to analyze, modify, or extract information from signals. It is widely used in real-world applications, including:

- Digital filtering and frequency-spectrum analysis.
- Frequency-domain processing, such as noise removal and image enhancement. For example, frequency-domain techniques can be used to reduce the appearance of thin clouds in satellite images.
- Digital audio crossover systems, speech coding, and speech compression.
- Interference and noise cancelation in electrocardiography (ECG) signals.

1.3 Continuous-Time vs. Discrete-Time Signals

Signals can be classified according to the domain of their independent variable. In many applications, the independent variable is time.

- **Continuous-Time Signals:** The independent variable can take any value within a continuous interval. A continuous-time signal is commonly denoted by $x(t)$, where $t \in \mathbb{R}$ represents continuous time. The signal is defined at every time instant, not only at selected time points. Examples include an analog audio waveform, a continuously varying voltage, and atmospheric pressure as a function of time.

A continuous-time signal does not necessarily have to be continuous in amplitude, such as a rect-angle pulse signal. It may contain discontinuities at certain time instants. Therefore, the term *continuous-time* refers to the continuous domain of the independent variable, rather than to the continuity of the signal value.

To determine whether a signal $x(t)$ is continuous at $t = t_1$, consider a small positive number $\epsilon > 0$ and examine the signal values immediately to the left and right of t_1 .

The left-hand limit is defined as

$$x(t_1^-) = \lim_{\epsilon \rightarrow 0^+} x(t_1 - \epsilon),$$

and the right-hand limit is defined as

$$x(t_1^+) = \lim_{\epsilon \rightarrow 0^+} x(t_1 + \epsilon).$$

The signal $x(t)$ is continuous at $t = t_1$ if

$$x(t_1^-) = x(t_1) = x(t_1^+).$$

In other words, the signal value approaching t_1 from the left, the actual value at t_1 , and the signal value approaching t_1 from the right must all be equal.

- **Discrete-Time Signals:** The independent variable takes only discrete values, which are commonly represented by an integer index n . A discrete-time signal is denoted by $x[n]$, where $n \in \mathbb{Z}$. The signal is defined only at specific time instants or sample indices.

A discrete-time signal may be obtained by sampling a continuous-time signal. If a continuous-time signal $x(t)$ is sampled every T_s seconds, the resulting discrete-time signal is

$$x[n] = x(nT_s),$$

where $T_s > 0$ is the fixed positive real number sampling period and n is an integer. Examples include digital audio samples, daily sales records, the amount of a loan payment made each month, and the number of children born on each day.

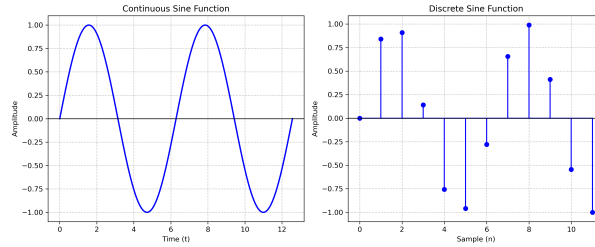


Figure 1: Comparison between a continuous-time sinusoidal signal and its discrete-time representation. The continuous-time signal is defined at every time instant, whereas the discrete-time signal is represented only at discrete sampling instants.

2 Continuous-Time Signals

2.1 Periodic vs. Aperiodic Signals

A continuous-time signal $x(t)$ is periodic if there exists a positive constant T such that

$$x(t + T) = x(t)$$

for every value of t . The value $T > 0$ is called a period of the signal. If the signal has a smallest positive period, it is called the **fundamental period** and is denoted by T_0 .

If T_0 is the fundamental period, then the signal also repeats after every integer multiple of T_0 . Therefore,

$$x(t + nT_0) = x(t),$$

where n is any integer. The values $2T_0$, $3T_0$, and so on are also periods, but they are not the fundamental period because they are not the smallest positive periods.

A signal that does not repeat after any positive time interval is called **aperiodic**.

- A standard example of a periodic signal is the sinusoidal function

$$x(t) = A \sin(\omega_0 t + \phi),$$

where A is the amplitude, ω_0 is the angular frequency in radians per second, and ϕ is the phase angle.

- The fundamental period of a sinusoidal signal is

$$T_0 = \frac{2\pi}{|\omega_0|},$$

provided that $\omega_0 \neq 0$.

- The ordinary frequency f_0 , measured in hertz (Hz), is related to the angular frequency by

$$\omega_0 = 2\pi f_0.$$

Therefore,

$$T_0 = \frac{1}{f_0}.$$

- If a signal is periodic with a fundamental period T_0 , then $2T_0$, $3T_0$, and other positive integer multiples of T_0 are also periods.
- Consider two periodic signals with fundamental periods T_1 and T_2 . Their linear combination

$$z(t) = ax(t) + by(t)$$

is periodic if the ratio of their fundamental periods is rational:

$$\frac{T_1}{T_2} \in \mathbb{Q}.$$

Equivalently, there must exist positive integers k and l such that

$$kT_1 = lT_2 = T.$$

The value T is a common period of the two signals.

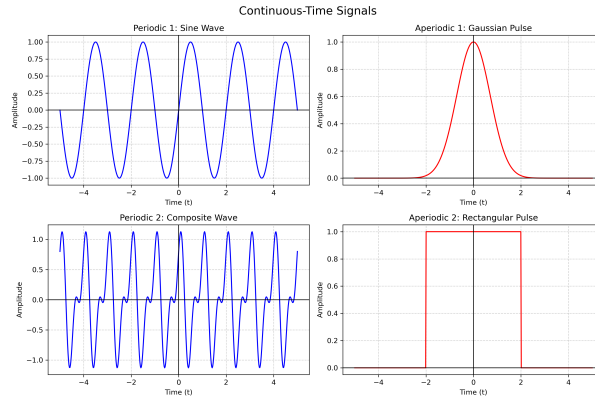


Figure 2: Comparison of continuous-time periodic signals(left) and aperiodic signals (right).

2.1.1 Complex Exponential Functions

Complex exponential functions are important in signal processing because they provide a compact representation of sinusoidal signals and simplify the analysis of linear systems.

A complex number can be written in rectangular form as

$$z = a + jb,$$

where a is the real part, b is the imaginary part, and $j = \sqrt{-1}$.

The symbol j is commonly used instead of i in electrical engineering because i is often used to represent electric current.

Euler's formula states that

$$e^{jx} = \cos(x) + j \sin(x).$$

Therefore, a complex exponential with phase θ can be written as

$$e^{j\theta} = \cos(\theta) + j \sin(\theta).$$

Similarly, a complex exponential as a function of time is

$$e^{j\omega t} = \cos(\omega t) + j \sin(\omega t),$$

where ω is the angular frequency in radians per second.

The complex exponential contains both a cosine component and a sine component:

$$\underbrace{\cos(\omega t)}_{\text{real part}} + j \underbrace{\sin(\omega t)}_{\text{imaginary part}}.$$

The complex exponential $e^{j\omega t}$ is periodic when $\omega \neq 0$, and its fundamental period is $T_0 = \frac{2\pi}{|\omega|}$.

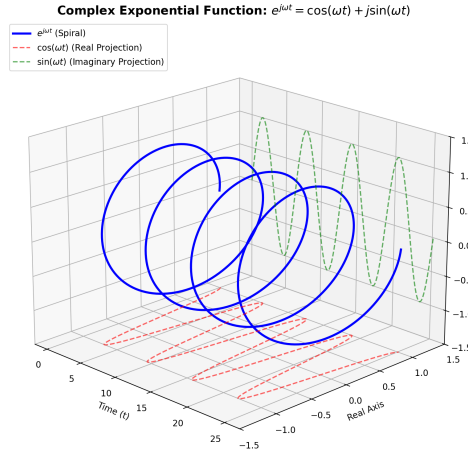


Figure 3: A 3D visualization of the complex exponential function $e^{j\omega t}$. The blue spiral represents the complex signal evolving over time. The red dashed line illustrates its projection onto the real axis, yielding the cosine function ($\cos(\omega t)$). The green dashed line illustrates its projection onto the imaginary axis, yielding the sine function ($\sin(\omega t)$).

2.1.2 Fundamental Frequency and Harmonics

The **fundamental frequency** is the lowest positive frequency of a periodic signal. It is denoted by f_0 .

A **harmonic** is a sinusoidal component whose frequency is an integer multiple of the fundamental frequency.

The n th harmonic has frequency $f_n = n f_0$, and angular frequency $\omega_n = n \omega_0$, where $n = 1, 2, 3, \dots$.

The first harmonic is the fundamental frequency itself. Higher harmonics are integer multiples of the fundamental frequency.

2.2 Energy vs. Power Signals

Signals can be classified based on their energy and power characteristics:

- **Energy Signal:** A signal with finite total energy ($0 < E < \infty$) and zero average power. The continuous-time energy is defined as:

$$E = \lim_{L \rightarrow \infty} \int_{-L}^L |x(t)|^2 dt$$

- **Power Signal:** A signal with finite average power ($0 < P < \infty$) and infinite energy. The average power is given by:

$$P = \lim_{L \rightarrow \infty} \left(\frac{1}{2L} \int_{-L}^L |x(t)|^2 dt \right)$$

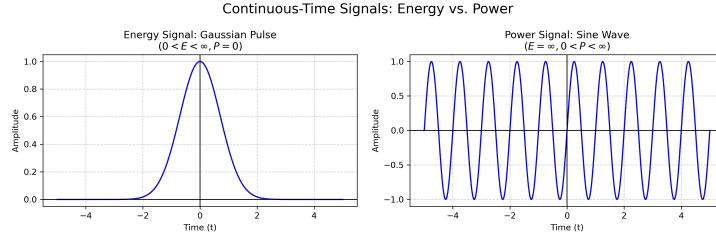


Figure 4: Comparison of continuous-time signals: an energy signal with finite energy and zero average power (left), and a power signal with infinite energy and finite average power (right).

2.3 Transformations of the Independent Variable

- **Shifting:** $x(t - t_0)$ represents a time-shifted version of $x(t)$. If $t_0 > 0$, the signal is delayed, and if $t_0 < 0$, the signal is advanced.
- **Reflection:** $x(-t)$ denotes the reflection of the signal about $t = 0$, analogous to playing an audio recording backward.
- **Time-Scaling:** $x(\eta t)$ scales the signal by a parameter η . It compresses the signal if $|\eta| > 1$ and expands it if $|\eta| < 1$. Playing the recording at a faster speed is analogous to compressing the signal in time, while playing it at a slower speed is analogous to expanding the signal in time. Thus, $x(2t)$ corresponds to a signal that changes twice as quickly and has half the original duration, whereas $x(t/2)$ changes more slowly and has twice the original duration.

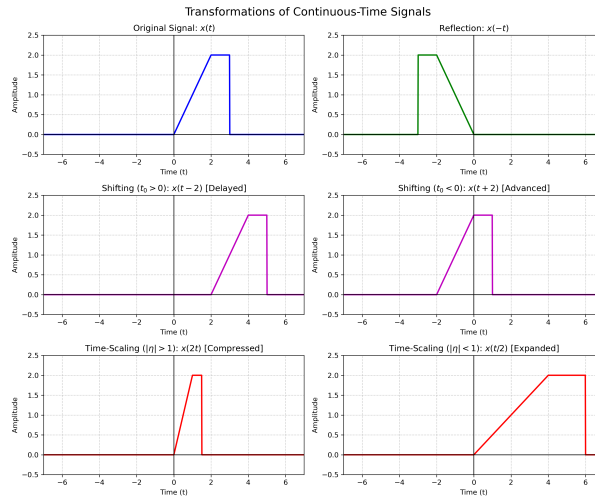


Figure 5: Transformations of a continuous-time signal $x(t)$. The unique asymmetric shape illustrates how reflection, shifting (delay/advance), and time-scaling (compression/expansion) modify the signal.

2.4 Even and Odd Signals

A signal can be classified as **even**, **odd**, or neither, according to its symmetry with respect to the origin.

2.4.1 Even Signals

A signal $x(t)$ is even if it satisfies $x(-t) = x(t)$ for every value of t .

An even signal is symmetric about the vertical axis $t = 0$. In other words, the signal has the same value at t and $-t$.

Common examples of even signals include:

- **Cosine signal:** $x(t) = \cos(\omega t)$
- **Squared signal:** $x(t) = t^2$
- **Absolute-value signal:** $x(t) = |t|$
- **Gaussian signal:** $x(t) = e^{-t^2}$

2.4.2 Odd Signals

A signal $x(t)$ is odd if it satisfies $x(-t) = -x(t)$ for every value of t .

An odd signal has origin symmetry. If the signal is rotated by 180° about the origin, its graph remains unchanged.

For an odd signal, $x(0) = 0$, provided that the signal is defined at $t = 0$.

Common examples of odd signals include:

- **Sine signal:** $x(t) = \sin(\omega t)$
- **Linear signal:** $x(t) = t$
- **Cubic signal:** $x(t) = t^3$
- **Signum function:** $x(t) = \text{sgn}(t)$

2.4.3 Even-Odd Decomposition

Any signal $x(t)$ can be uniquely decomposed into an even component $x_e(t)$ and an odd component $x_o(t)$:

$$x(t) = x_e(t) + x_o(t),$$

where

$$x_e(t) = \frac{1}{2} [x(t) + x(-t)]$$

is the even part, and

$$x_o(t) = \frac{1}{2} [x(t) - x(-t)]$$

is the odd part.

The even component is obtained by averaging the original signal and its time-reversed version:

$$x_e(t) = \frac{x(t) + x(-t)}{2}.$$

The odd component is obtained by taking half of their difference:

$$x_o(t) = \frac{x(t) - x(-t)}{2}.$$

2.5 Elementary Signals

Fundamental elementary signals include:

- **Unit Step Function:** The unit step function, denoted by $u(t)$, represents a signal that changes from zero to one at $t = 0$:

$$u(t) = \begin{cases} 0, & t < 0, \\ 1, & t > 0. \end{cases}$$

- **Ramp Function:** The unit ramp function is defined as

$$r(t) = \begin{cases} 0, & t < 0, \\ t, & t \geq 0. \end{cases}$$

This is obtained by integrating the unit step function.

- **Unit Impulse Function (Dirac Delta):** $\delta(t)$ acts as a spike at the origin where $\delta(0) \rightarrow \infty$ and $\delta(t) = 0$ for $t \neq 0$. The area under the impulse is $\int_{-\infty}^{\infty} \delta(t) dt = 1$. The delta function is the formal derivative of the unit step function.
- **Sampling Function:** The sampling function is defined as

$$\text{Sinc}(x) = \frac{\sin x}{x}, \quad x \neq 0.$$

At $x = 0$, the expression $\sin x/x$ has the indeterminate form $0/0$. However, its limit is

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

Therefore, the function is defined at the origin as $\text{Sinc}(0) = 1$ and its amplitude decreases as $|x|$ increases.

- **Rectangular Pulse Function:** The rectangular pulse function, denoted by $\text{rect}(t)$. Rectangular pulse signal extended from $-a$ to a with an amplitude A can be written as

$$A \text{rect}\left(\frac{t}{2a}\right) = A(u(t+a) - u(t-a))$$

- **Signum Function:** The signum function, denoted by $\text{sgn}(t)$, indicates whether its input is positive, negative, or zero:

$$\text{sgn}(t) = \begin{cases} -1, & t < 0, \\ 0, & t = 0, \\ 1, & t > 0. \end{cases}$$

It is related to the unit step function by

$$\text{sgn}(t) = 2u(t) - 1.$$

3 Discrete-Time Signals

3.1 Energy vs. Power Signals

For a discrete-time signal $x[n]$ and some integer $N > 0$:

- **Energy (E):** $E = \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x[n]|^2$.
- **Average Power (P):** $P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$.

The sequence $x[n]$ is classified as an energy signal if E is finite. It is considered a power signal if E is not finite, but P is finite.

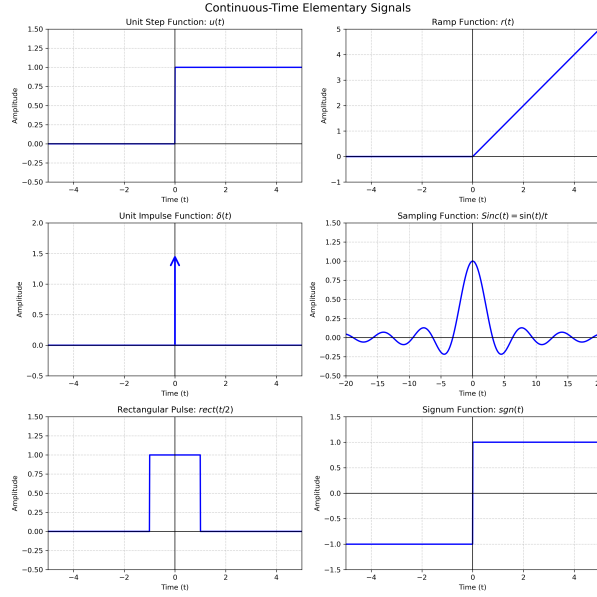


Figure 6: Continuous-time elementary signals: Unit Step, Ramp, Unit Impulse (Dirac Delta), Sampling Function, Rectangular Pulse, and Signum Function.

3.2 Periodic vs. Aperiodic Signals

- A signal $x[n]$ is periodic if, for some integer $N > 0$, it satisfies $x[n + N] = x[n]$ for all n .
- The smallest value of N that satisfies this relation is the fundamental period of the signal. If no such integer N exists, $x[n]$ is an aperiodic signal.

3.3 Transformations of the Independent Variable

Discrete-time signals are defined only at integer-valued indices. Therefore, transformations of the independent variable must preserve integer indices or be interpreted using an appropriate discrete-time operation.

3.3.1 Shifting

For an integer value k , the sequence

$$y[n] = x[n - k]$$

is a shifted version of $x[n]$.

- If $k > 0$, $x[n - k]$ is shifted to the right, or delayed, by k samples.
- If $k < 0$, $x[n - k] = x[n + |k|]$ is shifted to the left, or advanced, by $|k|$ samples.

3.3.2 Reflection

The operation

$$y[n] = x[-n]$$

represents time reversal, or reflection, of the sequence $x[n]$ about the origin $n = 0$.

3.3.3 Time Scaling

Time scaling in discrete-time signals must be interpreted carefully because a discrete-time sequence is defined only for integer values of the index n .

Compression For an integer $M > 1$,

$$y[n] = x[Mn]$$

is called compression or downsampling by a factor of M . Since Mn is always an integer, the expression is well defined.

The output sequence retains only every M th sample of the original sequence. For example,

$$y[n] = x[2n]$$

retains the samples

$$\dots, x[-4], x[-2], x[0], x[2], x[4], \dots$$

and discards the samples at odd indices.

Thus, $x[2n]$ does not simply “squeeze” the plotted sequence in the same way as continuous-time scaling. It performs sample selection and reduces the number of samples.

Expansion The expression

$$x\left[\frac{n}{M}\right]$$

cannot be directly evaluated for all integer values of n because n/M is not an integer unless n is divisible by M .

For example, when $M = 2$,

$$x\left[\frac{n}{2}\right]$$

is defined directly only when n is even. At odd values of n , the index is not an integer:

$$\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$$

Therefore, discrete-time expansion is usually defined by inserting additional samples between the original samples. For expansion by a factor of M ,

For example, expansion by a factor of two gives

$$y[2m] = x[m],$$

and

$$y[2m + 1] = 0.$$

Thus, the original samples are retained at even indices, while zeros are inserted at odd indices. This operation is also called upsampling by a factor of two.

3.4 Elementary Signals

- **Unit Impulse Function:** Defined in discrete time as $\delta[n] = 1$ for $n = 0$, and 0 for $n \neq 0$. While somewhat similar to the continuous-time impulse $\delta(t)$, the magnitude of the discrete impulse is always finite.
- **Unit-Step Sequence:** Defined as $u[n] = 1$ for $n \geq 0$, and 0 for $n < 0$.
- **Relationship:** The first difference of the unit-step function yields the discrete impulse: $u[n] - u[n - 1] = \delta[n]$.
- **Other Properties:** Discrete-time impulse and step functions exhibit Several important mathematical properties.
 - Running sum of the impulse: $\sum_{k=-\infty}^n \delta[k] = u[n]$.
 - Alternatively: $\sum_{k=0}^{\infty} \delta[n - k] = u[n]$.
 - Sampling property: $x[n]\delta[n - k] = x[k]\delta[n - k]$.
 - General representation form of a sequence: $x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n - k]$.

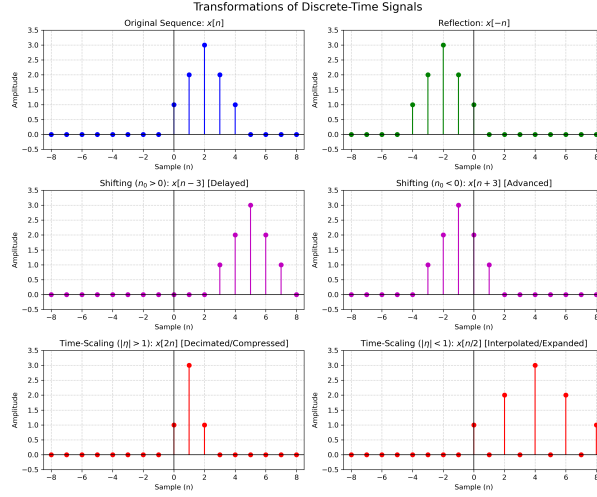


Figure 7: Transformations of a discrete-time signal $x[n]$. Note that time-scaling in discrete time behaves uniquely: $x[2n]$ results in decimation (loss of intermediate samples), while $x[n/2]$ expands the sequence by inserting zeros at odd indices.

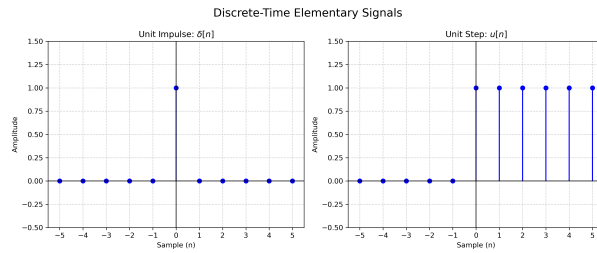


Figure 8: Discrete-time elementary signals: the discrete unit impulse sequence $\delta[n]$ (left) and the discrete unit step sequence $u[n]$ (right).